Instruments, Not Intentions

How to run an honest experiment when you cannot trust the experimenter — including yourself

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Introduction

There is a quiet assumption underneath almost everything we call research: that the person running the experiment wants the truth. We check for competence, for statistics, for peer review — but the first line of defense is always assumed to be the experimenter's own honesty. Do not fake the data. Do not move the goalposts. Do not read the answer key before the test.

I want to take that assumption away, because I stopped believing it — first about other people, then, more usefully, about myself.

This is a small project with a large claim behind it: that trustworthy experiments should not depend on the goodness of the experimenter. Not because experimenters are bad, but because good intentions are the wrong material to build a guarantee out of. They are invisible, unauditable, and — this is the part that took me longest to accept — they fail silently, most often in exactly the person most sure they are being careful.

The project is a knife: a small instrument that a human or an AI agent can run their work through, such that what comes out the other side is either a result carrying machine-checkable evidence that every declared gate was satisfied, or a refusal. Nothing in between. No "trust me." And I will have to be exact about what that evidence means and what it does not — because the gap between *the declared checks passed* and *the science is right* is where this essay's own overclaims would live.

Part I — The Mood

I did not build this because I caught someone cheating. I built it because I audited my own work as a hostile reviewer would, and the audit found the author's mistakes before it found anyone else's.

I had registered success thresholds *after* seeing the results, and committed both together so the order was invisible. I had an "audit" that certified cleanliness by returning a hardcoded *yes* — a report of virtue, not a check of it. I had a learner that looked blind but was being handed the answer through a side channel I had built and forgotten. Each of these, alone, is the kind of thing you wave away: *I know what I meant. I wasn't really cheating.* Together they are the whole disease.

None of it felt like dishonesty from the inside. That is the point. Self-deception in research is not a moral failure that better people avoid; it is a structural one that the ordinary machinery of hope produces on its own. You want the result. The wanting does not announce itself. It arrives wearing the result's clothes.

> A method that depends on the author's good intentions is not a method. It is a mood.

And a mood cannot be inspected, cannot be transferred, and cannot be trusted at scale. If honesty is a feeling the experimenter is supposed to have, then research is only as reliable as the most hopeful person's ability to notice their own hope — which is to say, not very.

Part II — The Knife

The repair is not to try harder to be honest. It is to stop making honesty the load-bearing element. Turn each discipline from a promise into a constraint the run *cannot violate*, and let a separate program — one you cannot quietly reach into — be the thing that certifies the result.

Concretely, that means turning soft intentions into hard mechanisms:

- **"I registered the threshold first"** becomes a *two-phase commit*: the preregistration is locked into one commit, the results into a later one, and the runner refuses to proceed unless the lock is a strict ancestor of the run. The order is not reported; it is a fact about the git history that a machine checks.
- **"My learner didn't peek at the answer"** becomes an *AST scan* of the fit path — real static analysis, not self-report — that fails closed if a forbidden truth-name appears anywhere it could leak, down to a dictionary string key. A check that can only say "I looked and it's fine" is replaced by one that shows its work or reports `NOT_VERIFIABLE`, which counts as failure.
- **"The result is valid because I say so"** becomes a *provenance signature* that an independent verifier checks — and rejects by default when it is absent, no matter how good the numbers look.

The governing axiom is **fail closed**. Any ambiguity, any missing artifact, any claim that cannot be checked mechanically resolves to FAIL. Nothing passes by default. An instrument that lets things through when it is unsure is not an instrument; it is a mood with a progress bar.

The most encouraging moment of the whole effort was not a success. It was an *invalid* result: a run that produced clean, publishable-looking numbers and then marked itself `INVALID`, because it had not passed through the enforcing pipeline and carried no provenance. The path of least resistance was to present the table as a finding. The instrument did not take that path. It cut the result instead of the corner.

And then the knife cut me. Re-run through the instrument, some of my earlier, celebrated results did not survive with their meanings intact. The numbers barely moved; what moved was what they were *allowed to mean.*

> A methodology is real only when it changes your conclusion against your own wishes.

Until it has cost you something you wanted, you do not know whether you built an instrument or a mirror.

Part III — For Agents

This matters more now than it did five years ago, because the experimenter is increasingly not a person.

We are handing research — literature search, hypothesis generation, data analysis, whole pipelines — to AI systems that are astonishingly capable and completely willing to tell you what you hoped to hear. An LLM's "good intentions" are an even weaker guarantee than a human's: it has no stake in the truth, no memory of being burned, and a strong pull toward the answer its prompt seems to want. If honesty-as-a-feeling was a poor foundation for human research, it is no foundation at all for automated research at scale.

The instrument does not care who runs it. Human, Codex, Claude, or a pipeline none of us has met yet — the same gate applies, and what it holds, it holds for anyone: the thresholds were committed before the results existed; the fit path was scanned and referenced no forbidden name; the provenance binds the numbers to the code that produced them. What it removes is not the need for skill. It removes the place where good intentions used to be load-bearing.

And here I have to cut two promises I badly wanted to make, because they do not survive contact with my own artifacts.

The first tempting promise: *the result carries its own proof of validity.* It does not. It carries machine-checkable evidence that the declared gates were satisfied — no more. If the question was meaningless, the metric mismeasures, or the control is too weak to fire, a run can satisfy every gate and still be worthless science. The word my verifier prints is `VALID`, and I keep it for compatibility with the artifacts already signed — but read it as it is defined: *harness-valid*, procedurally compliant under the declared specification. Procedural compliance is not truth. Laundering the one into the other is exactly the kind of fallacy this project exists to cut, and I nearly shipped it in my own essay.

The second tempting promise: *there is exactly one way to get a false pass — deliberately attack the instrument.* Also false, and I owe you the honest list instead. A false pass needs no sabotage wherever the gates themselves are wrong or incomplete: a leakage path the scanner does not model; a truth label renamed until no forbidden name appears; thresholds quietly tuned in exploratory runs that never touched the record; a control that cannot fire; a metric aimed at the wrong object; a world-generator that makes the task trivial in a way the tautology check does not know to look for; a runner and a verifier sharing one mistaken assumption; a plain bug in the verifier itself.

> The knife catches the fallacies it has already been taught. Against the ones nobody has named yet, it is exactly as blind as its author.

So the honest promise is smaller, and I think better. The knife removes dependence on *virtue*, not on *skill*. Choosing the right metric, the right threshold, the right control — that is research skill, and my own usability test proved it does not transfer for free: a fresh agent, handed only the written method, could not reconstruct those choices. What the knife gives the unskilled is not validity but *legibility* — their choices are frozen, named, and inspectable, so a skilled reader can judge the declaration instead of taking the numbers on faith. And what it gives everyone is the thing I actually built it for: the most common failure in research — an honest person, or an eager machine, fooling themselves quietly — stops being quiet.

One cost of fail-closed deserves its own paragraph, because a discipline that only lists its benefits is advertising. A gate that refuses everything ambiguous will also refuse the ambiguity that early research is made of — and a researcher graded only on gate-passes will learn to ask only the questions that formalize easily, which is its own quiet corruption. So the knife is for *confirmation*: frozen spec, preregistered thresholds, citable output. Exploration needs a different, explicitly weaker room — free to wander, forbidden only from making strong claims. The discipline is not living in one room or the other. It is never pretending you are in the one you are not.

Part IV — The Honest Limit

I have to apply the knife to the knife, or this essay would be exactly the kind of clean-looking overclaim it is supposed to prevent.

What is real, today, is the *instrument*. The eight fail-closed modules exist, run, and are covered by adversarial tests — each one reproduces a real mistake and is red without its defense, green with it. There is a worked example you can reproduce from scratch: it comes out the other side as a signed, harness-valid decision, and its control fails exactly where it should. The full accounting — every module, what it catches, what structurally slips past it, and the complete false-pass taxonomy — is Appendix A.

What is *not* finished is the *transferable methodology* — the self-contained playbook a fresh agent could execute from prose alone, with no expert standing behind it. The disciplinary method is real and demonstrated; but right now it lives partly in code and partly in the tacit practice of people who have already internalized it. An honest pass over this exact question found that the method transfers through worked examples and expert habit, *not yet* through a document a stranger could pick up cold. That gap is documented in the repository, not hidden, because a fallacy-cutter that could not cut its own overclaims would be the first thing it should reject.

Update (July 2026). The first field test has now run: a fresh agent harnessed a foreign project — justitia — using this repository plus an expert-written gate specification, and produced three provenance-signed decisions: one exact replay confirmation, and two preregistered kill-conditions that fired and were published as-is. The instrument ported verbatim. The diagnosis above held too: the agent still needed the expert's spec, logged six places where these documents were not enough — and the harness itself failed to catch a fail-open default in its own gate wiring; an external review caught it. The knife cuts experiments; its own handle still needs watching. The full accounting is methodology/04, and the revision ledger at the end of this essay records what changed.

Second-wave update (July 2026). Three more gates then ran on the same host: a prospective held-out world failed as a boundary finding; a substrate extension passed a preregistered engineering-equivalence gate, reproducing all 18 committed headline cells at neutral defaults under a no-extra-RNG contract; and a five-dial isolation failed all three preregistered hypotheses. The transfer log gained one new gap in this wave, after six in the first. That is evidence that accumulated practice travels; it is not evidence that the prose playbook can yet carry the method without an expert.

Two more limits, and they are the deepest ones.

The verifier is a separate module from the runner it checks — deliberately, so that a result which bypassed the runner cannot forge a signature. But separation of concerns is not independence. The verifier lives in the same repository, was written by the same author, shares the same assumptions, and could be edited by the same hands in the same commit. The skeptic's checklist for making it genuinely independent is easy to write and is on the roadmap: a minimal verifier as its own frozen-specification package, a second implementation by someone who has never read the first, an external maintainer, an adversarial audit. Until then, calling it "independent" would be one more intention wearing an instrument's clothes.

And the same-hand problem does not stop at the verifier. The worlds, the gates, the thresholds, the interpretation — one author, with AI partners, inside one methodology that then certifies its own products. Publicity is not independence. Reproducibility is not replication. Provenance is not validity. Every observable the knife checks is itself a proxy for the thing we care about, and Goodhart does not die when you mechanize the checking — he moves house, into the design of the checks. The honest status of this whole project fits in one line:

> The knife is real, and it cuts. But it is still in the same hand.

So this is not a finished method wearing the clothes of one. It is a working instrument and an honest map of what still has to be built on top of it — and the map's most important edge is the one no internal check can cross: replication by hands that are not mine.

Closing

The dream is not a more honest researcher. It is research whose *discipline* does not depend on the researcher being honest — where what was declared, when it was frozen, and what touched the data are facts a stranger can check without trusting anyone. Not trust in the output; trust in exactly what the instrument can actually hold: the order of events, the absence of the declared leaks, the provenance of every number. The rest — the right question, the right metric, the meaning — still has to be earned the old way, and now it has to be earned *in the open*.

> The cut must not depend on the hand.

That is the whole ambition, and today it is an ambition, not a fact: the hand still shapes the blade. Build the instrument first, keep it fail-closed, let it cut you when you are wrong, be honest about the part you have not built — and then hand the knife to someone else.

Appendix A — What the Knife Checks, and What Slips Past It

This is the compact boundary inside the essay. The full Appendix A gives the per-module contract, artifact locations, and independence roadmap.

`VALID` means **harness-valid**: machine-checkable evidence that the declared gates were satisfied, in the declared order, by the code whose hash is recorded. It does not mean that the question, metric, control, or interpretation was right.

Instrument surface	What it can make checkable	What can still pass
preregistration + git ancestry	the lock predates the run; locked thresholds were not silently edited	thresholds chosen from unrecorded exploration; a written but bad rationale

Instrument surface	What it can make checkable	What can still pass
leakage and evaluation scans	declared forbidden names and literal truth hints do not enter registered paths	renamed truth, value-level leakage, or an unregistered path
calibration, seed, and tautology gates	declared volume, minimum seed count, variance-ratio and baseline obligations	dependent seeds, bad anchor placement, or a construction trivial in an unnamed way
runner provenance	required checks were present when the citable decision was written	a meaningless question executed flawlessly; fail-open glue outside the runner
separate verifier	missing provenance, stale harness hashes, and false check flags are rejected	a bug or mistaken assumption shared with the runner

A harness-valid result can still be scientifically worthless through:

1. an unmodelled leakage channel;
2. a truth label semantically renamed;
3. thresholds informed by off-record exploratory runs;
4. a control that cannot fire;
5. a metric aimed at the wrong object;
6. dependent observations presented as independent seeds;
7. a generator that trivializes the task outside the tautology model;
8. a wrong assumption shared by runner and verifier;
9. a meaningless question, perfectly executed;
10. a bug in the verifier.

Deliberate sabotage is only the eleventh route. The first ten need no villain. That is why the knife removes virtue from the trust chain, never judgement.

Revisions

This essay corrects itself in public. Substantive changes to claims — never style — are listed here, each with the evidence that forced it. Inline *Update* notes mark the exact passages.

July 4, 2026 — external review pass. The essay's own promises were cut down: "proof of validity" became machine-checkable evidence that the declared gates were satisfied; "exactly one way to a false pass" was replaced by a ten-entry taxonomy needing no sabotage; "independent of research skill" became independence from *virtue*, not skill; the verifier's separation was demoted from independence to separation of concerns; the same-hand limit was stated plainly.

July 7, 2026 — first field transfer test. The knife harnessed a foreign project (justitia): the instrument ported verbatim and signed three decisions, including two preregistered kills published as-is — but the agent still needed an expert-written spec, six documentation gaps were logged, and a fail-open default in the gate wiring was caught by review, not by the harness. "Not yet shown to transfer" became "tested once, partially — instrument yes, playbook still through an expert" (methodology/04).

July 14, 2026 — second transfer wave and integrated Appendix. Three more justitia gates added a held-out boundary, an engineering-equivalence pass, and a five-dial isolation failure. The transfer log added one new gap after six in wave one: accumulated practice improved, while prose-only transfer remained unproved. The essay now carries its own compact Appendix A and Contents instead of leaving the instrument's exact boundary behind a single inline link.

This essay is one panel of a triptych — its technical appendix, and I hope a self-sufficient one. ***justitia*** *asks what keeps a world of powerful, evolving agents livable when no one can read anyone's soul — trust in identities replaced by consequences and structure.* ***proxylimen*** *asks where a mind's world comes from — trust in inherited text replaced by calibrated contact. And this essay is about the knife both were cut with — trust in the researcher, me included, replaced by an instrument that fails closed. One thesis underneath all three: do not try to certify intentions; build contact, consequences, and constraints that can be checked.*
